

Semi-Deviation: Overview, Formulas, History

 [investopedia.com/terms/s/semideviation.asp](https://www.investopedia.com/terms/s/semideviation.asp)

What Is Semi-Deviation?

Semi-deviation is a method of measuring the below-mean fluctuations in the returns on investment.

Semi-deviation will reveal the worst-case performance to be expected from a risky investment.

Semi-deviation is an alternative measurement to standard deviation or variance. However, unlike those measures, semi-deviation looks only at negative price fluctuations. Thus, semi-deviation is most often used to evaluate the downside risk of an investment.

Understanding Semi-Deviation

In investing, semi-deviation is used to measure the dispersion of an asset's price from an observed mean or target value. In this sense, dispersion means the extent of variation from the mean price.

Key Takeaways

- Semi-deviation is an alternative to the standard deviation for measuring an asset's degree of risk.
- Semi-deviation measures only the below-mean, or negative, fluctuations in an asset's price.
- This measurement tool is most often used to evaluate risky investments.

The point of the exercise is to determine the severity of the downside risk of an investment. The asset's semi-deviation number can then be compared to a benchmark number, such as an index, to see if it is more or less risky than other potential investments.

The formula for semi-deviation is:

$$\text{Semi-deviation} = \sqrt{n-1 \times \sum_{rt < \text{Average}} (\text{Average} - rt)^2}$$



where: n = the total number of observations below the mean

rt = the observed value
 average = the mean or target value of a data set

An investor's entire portfolio could be evaluated according to the semi-deviation in the performance of its assets. Put bluntly, this will show the worst-case performance that can be expected from a portfolio, compared to the losses in an index or whatever comparable

is selected.

History of Semi-Deviation in Portfolio Theory

Semi-deviation was introduced in the 1950s specifically to help investors manage risky portfolios. Its development is credited to two leaders in modern portfolio theory.

- Harry Markowitz demonstrated how to exploit the averages, variances, and covariances of the return distributions of assets of a portfolio in order to compute an efficient frontier on which every portfolio achieves the expected return for a given variance or minimizes the variance for a given expected return. In Markowitz' explanation, a utility function, defining the investor's sensitivity to changing wealth and risk, is used to pick an appropriate portfolio on the statistical border.
- A.D. Roy, meanwhile, used semi-deviation to determine the optimum trade-off of risk to return. He didn't believe it was feasible to model the sensitivity to risk of a human being with a utility function. Instead, he assumed that investors would want the investment with the smallest likelihood of coming in below a disaster level. Understanding the wisdom of this claim, Markowitz realized two very important principles: Downside risk is relevant for any investor, and return distributions might be skewed, or not symmetrically distributed, in practice. As such, Markowitz recommended using a variability measure, which he called a semivariance, as it only takes into account a subset of the return distribution.